

YaPcore: YaP-Folia Game Authority, Slim Edge Chassis, Native Network Stack, and First-Party Plugin Suite

YapLabs Technical Whitepaper

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Prefer plain English? See [YAPCORE_WHITEPAPER_PLAIN_ENGLISH.md](#).
Operator rundown: [YAPCORE_WHITEPAPER.md](#).

Abstract

Minecraft-class game servers traditionally serialize world mutation, plugin callbacks, and much of network I/O onto a single “main” thread, trading simplicity for latency under concurrent load. **YaPcore** ships as a three-layer product:

1. **YaP-Folia** — YapLabs’ maintained fork of PaperMC Folia **26.2** — owns regionized game tick (`lib/yap-folia-26.2.jar` , `folia-jar-source=build`).
2. **YapEngine** — a slim chassis in the YaPcore parent process — owns the public edge: watchdog, Netty traffic/sequencing, Compatibility Bridge, UI sandboxes, and heavy I/O workers (**not** world tick).
3. **YaP Link** — a first-party Velocity-class proxy (`0.6.0-phase6`) — fronts multi-backend networks.

A **SequenceToken** model orders work across chassis streams. Folia-aware first-party plugins use an explicit **SYNC / HEAVY / UI** contract via [YapSched](#) . The product ships a **CORE+NETWORK** plugin suite (permissions, chat, moderation, playerdata/economy, protect, world, regions, map, ...) and an opt-in **GAMEPLAY** tier (vehicles, stacker, knobs, and a full **MMO** stack M0–M7). Dual-stack **Java TCP + Bedrock UDP** is first-party (no Via*/Geyser jars). Stock Fill Folia and Paper + Phase 3 spatial tick remain **legacy / bench** paths only.

This paper describes architecture, concurrency invariants, networking/crossplay, the shipped plugin and data plane, evaluation methodology, and honest product status as of September 2026.

Keywords: game server concurrency; Folia fork; regionized tick; plugin compatibility; Minecraft protocol; dual-stack crossplay; generational ZGC; MariaDB; MMO progression.

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1. Introduction

1.1 Problem

Vanilla and Paper-derived Java Edition servers concentrate authoritative world updates on one thread (~20 TPS). Plugins that perform database or HTTP work on that thread stall ticks. Unsynchronized parallel mutation produces torn chunks, duplicate entities, and inventory races. Network operators still assemble **LuckPerms + EssentialsX + CoreProtect + WorldEdit + WorldGuard + Velocity + Geyser + ...** by hand. Bedrock clients require a second transport (UDP) while operators demand a single shared world.

1.2 Contribution

YaPcore contributes:

1. A **three-layer product stack**: YaP-Folia game tick + YapEngine edge/I/O chassis + YaP Link proxy.
2. A **maintained Folia fork** (branding, teleport transactions, optional hot-region budgets / partition) — [QUICK_START.md](#).

3. **SequenceToken** sequencing for ordered handoff across chassis threads — [YAPCORE_WHITEPAPER.md](#).
4. A **Compatibility Bridge** that stages legacy Bukkit mutations onto the game-core drain window (non-product Paper path; best-effort on Folia via Folia APIs + `YapSched`).
5. Dual-stack **Java TCP + Bedrock UDP** ingress with optional shared listen port — first-party code, not Via*/Geyser jars — [CROSSPLAY.md](#).
6. A **three-tier extension model**: Folia-aware plugins (`plugin.yml`), YaP plugins (`yap.yml`), and fine-tune modules (`module.yml`) — [PLUGINS.md](#) · [MODULES_AND_API.md](#).
7. A **shipped first-party plugin suite** that replaces the common DIY glue stack for ~90% of survival/network operators — §6.
8. An opt-in **RuneScape-style MMO progression stack** (skills, combat, crafting, quests, abilities, Bedrock UI) — [MMO_PHASES.md](#).

1.3 Non-goals

- **Stock Paper plugins on YaP-Folia** are unsupported (same reality as upstream Folia). Prefer Folia-aware jars or YaP natives — [PLUGIN_COMPAT_MATRIX.md](#).
- YaPcore is **not** a clean-room rewrite of Minecraft; it **forks Folia on purpose**.
- We do **not** claim “faster than Paper/Leaf on every workload.” Population cite: **fullcite** (100 active bots + fixtures) under the **ship Folia knob profile** (async-save, hopper budget, MSPT-gated entity/microtick budgets, subregion partition) — yapcore **−12.4%** vs stock Folia (`20260904TshipFc2`); knobs disclosed in bench JSON (`knob_*`). See [YAP_FOLIA_SOAK.md](#) . **250 keepalive = HOLD-ONLY**.
- Bedrock play-depth is **join/spawn + play-depth smoke green**; Wave 2 fidelity matrix (inventory/forms/G.33 heads) is in [CROSSPLAY.md](#) — Floodgate-only forms and anvil/smithing UIs are **Limited/Out**, not silent Partial.
- The Compatibility Bridge facade (non-game authority) remains **best-effort stubs** — [PAPER_API_COVERAGE.md](#).

1.4 Audience

Audience	Primary sections
Operators / network owners	§3, §6-10, §13
Plugin authors	§5-7, Appendix A
Systems / engine contributors	§3-4, §11-12
Non-technical readers	Plain English whitepaper

2. Related work

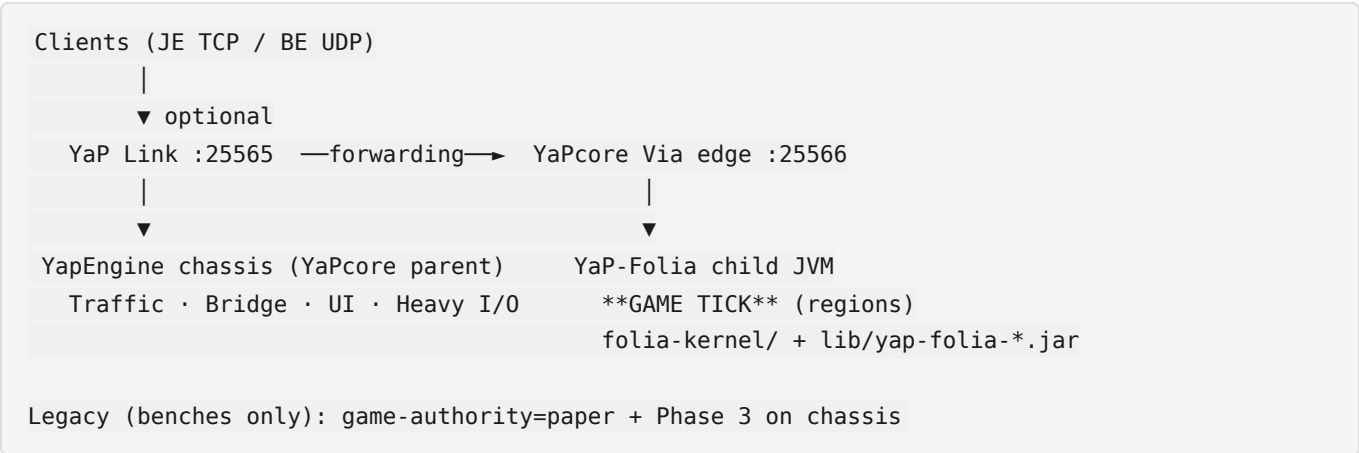
Paper / Purpur / Pufferfish / Leaf improve the classic Bukkit single-main-tick model (performance and/or knobs). **PaperMC Folia** introduces regionized multithreading for Bukkit-class servers. **YaPcore forks Folia** as **YaP-Folia** rather than shipping upstream Fill as the product jar. Peer Folia forks (e.g. Canvas) typically leave operators to assemble proxy + plugins themselves.

Netty-based proxies (Velocity; YaP Link) separate player routing from world authority. **Geyser / Via*** stacks provide dual-stack crossplay as separate jars; YaPcore embeds Via-class and Geyser-class code in chassis/ `crossplay` packages.

YaPcore sits as: **YaP-Folia’s game + deterministic YapEngine thread roles + first-party Link + shipped plugin/data plane**, rather than “DIY Folia + Velocity + ten community plugins.” Comparison matrix: [YAPCORE_WHITEPAPER.md](#).

3. Architecture

3.1 Three layers (product)



Layer	Game tick?	Role
YaP Link	No	Proxy JVM — multi-backend routing, compression, Link plugins (0.6.0-phase6)
YapEngine chassis	No	Edge, bridge, UI/Heavy I/O, telemetry, dual-stack gateway
YaP-Folia	Yes	Region thread pool in embedded JVM (lib/yap-folia-*.jar)

Default product properties:

```
game-authority=folia
folia-embed=true
```

```
folia-version=26.2
folia-jar-source=build
folia-teleport-transactions=true
paper-phase3-tick-bridge=false
paper-phase3-nms-tick=false
```

3.2 Chassis channel matrix (T1-16)

ID	Role	Responsibility (v2.0 / product)
1	Controller	Watchdog, recovery, process health
2	Traffic Cop	Ingress shaping, SequenceToken assignment
3-6	Worker quads	Sequenced bridge/plugin tasks; legacy Phase 3 NMS tick on Paper benches only
7	Chunk Sync DLM	Deferred lease / chunk ownership (Paper Phase 3 legacy)
8	Boundary Arbitrator	Cross-quadrant handoff (Paper Phase 3 legacy)
9	Compatibility Bridge	Legacy SYNC mutation queue → YaP-Folia region APIs
10-11	UI sandbox	Menu polish, click routing
12-15	Heavy I/O	DB, HTTP, files, proxy sync
16	Telemetry	Metrics / JFR hooks

See [YAPCORE_WHITEPAPER.md](#).

3.3 Sequencing

Each logical stream obtains a `SequenceToken` carrying a per-stream sequence and a global identifier. Strict ordered queues refuse out-of-order commits within a stream while allowing cross-stream parallelism — [YAPCORE_WHITEPAPER.md](#).

3.4 Spatial model

YaP-Folia indexes world interest by region and runs authoritative tick on a dynamic region thread pool. Optional YaP patches add teleport transactions (default **on**), entity tick budgets, async chunk save, scoreboard SWMR, and experimental subregion partition (default **off**) — §4. **YaPEngine chassis quads (T3-6)** route sequenced bridge/plugin work; they do **not** replace YaP-Folia game tick. Legacy Paper Phase 3 used quads + T7/T8 for interior NMS tick (**benches only**).

3.5 Memory & GC posture

Production launch scripts prefer **Generational ZGC** with optional **NUMA** pinning — [TUNE.md](#). **Java 25+** is required.

4. YaP-Folia fork

YaPcore does **not** ship stock PaperMC Folia as the product game jar. Upstream pin lives in `vendor/folia/UPSTREAM.lock`; ordered patches in `vendor/folia/patches/`.

Patch	Purpose	Default
0000	Branding → YaP-Folia	always
0001	Cross-region teleport PREPARE/COMMIT/CONFIRM	on
0010	Async chunk save (Moonrise flush off region thread)	on (ship)
0011	Scoreboard SWMR	on (ship)
0012 / 0023	Smart Mob AI tick budget (MSPT-gated)	400 (ship); 0=off
0013	Region pool metrics / microtick knobs	microtick 8 ms (ship)
0014 – 0019 / 0024	Subregion partition + harden	on (ship; hysteresis)
0022	Hopper BE transfer budget	64 (ship)

Build: `./scripts/build-yap-folia.sh` → `lib/yap-folia-26.2.jar`.

Docs: [YAP_FOLIA_SOAK.md](#) · [REAL_GAINS.md](#) · [QUICK_START.md](#).

Stock Folia fallback: `folia-jar-source=fetch` + `./scripts/fetch-folia.sh` (bench / comparison only).

5. Plugin concurrency contract

Pool	Allowed work	Forbidden
SYNC (region / entity / global via <code>YapSched</code>)	Block / inventory / teleport / world	Blocking DB / HTTP
HEAVY	JDBC, HTTP, disk, messaging	Direct block set / <code>openInventory</code> without hop
UI	Menu animation, polish	Authoritative world writes

Rules for every first-party jar:

- `folia-supported: true` in `plugin.yml`
- World/block/entity work via `com.yapcore.sched.YapSched` (or Folia region APIs)
- Shared persistence via **YaPDB** when MariaDB is required
- Public APIs registered on Bukkit `ServicesManager`

Operator layout: all jars drop into `plugins/` (symlinked into `folia-kernel/plugins`). Fine-tune packaging modules drop into `modules/`. See [PLUGINS.md](#) · [PLUGIN_COMPAT.md](#).

6. Shipped first-party plugins

Sources live under `yap-first-party/`. Install tiers:

Tier	Gradle	Audience
CORE+NETWORK	<code>gradle installProductDefaults</code>	Every release
GAMEPLAY	<code>gradle installGameplayDefaults</code> or <code>-PyapGameplay=true</code>	Opt-in survival / MMO
Both	<code>gradle</code> <code>installAllProductDefaults</code>	Full product box
Dist	<code>gradle assemblePluginDist</code>	<code>build/dist/yap-plugins/{core-network,gameplay,api,modules}/</code>

6.1 CORE + NETWORK (default product)

Jar	Plugin	Role
<code>yap-db.jar</code>	YaPDB	Shared MariaDB Hikari pool
<code>yap-perms.jar</code>	YaPPerms	Groups, tracks, prefixes (<code>/yapperm</code> , <code>/promote</code>)
<code>yap-playerdata.jar</code>	YaPPlayerData	Cross-server sync, auth, economy, homes/warps/kits/mail, chest shops , AH , claims
<code>yap-moderation.jar</code>	YaPModeration	Ban / mute / warn / kick + history
<code>yap-essentials.jar</code>	YaPEssentials	Essentials-class QoL (<code>/spawn</code> , <code>/tpa</code> , <code>/fly</code> , <code>/vanish</code> , ...)
<code>yap-chat.jar</code>	YaPChat	Channels, PM, filter, staff chat
<code>yap-packs.jar</code>	YaPPacks	Multi resource-pack push
<code>yap-floodgate.jar</code>	YaPFloodgate	Bedrock identity without Floodgate jar
<code>yap-placeholderapi.jar</code>	PlaceholderAPI	Clip-compatible PAPI (local expansions; eCloud out of scope by design)

Jar	Plugin	Role
yap-plugin-compat.jar	YaPPluginCompat	1.20-1.21 → 26.2 back-compat status
yap-pregen.jar	YaPPregen	Folia-safe chunk pre-generator
yap-folia-bridge.jar	YaPFoliaBridge	Folia surface / scheduler smoke
yap-protect.jar	YaPProtect	CoreProtect-class audit / rollback / restore
yap-world.jar	YaPWorld	FAWE-class edit (masks, brushes, schems) + world mgmt
yap-regions.jar	YaPRegions	WorldGuard-class cuboid flags
yap-npcs.jar	YaPNpcs	Quest NPCs + dialogue
yap-tab.jar	YaPTab	Tab list / header / footer / sidebar
yap-discord.jar	YaPDiscord	Discord webhooks + relay
yap-guard.jar	YaPGuard	Lightweight anti-cheat heuristics
yap-lagguard.jar	YaPLagGuard	Per-chunk lag governor
yap-map.jar	YaPMap	Web flat map (Leaflet + PNG tiles + live markers; 3D Stretch)
yap-factions.jar	YaPFactions	Factions overlay on playerdata claims
yap-bedrock-ui.jar	YaPBedrockUI	Bedrock FormService bridge

6.2 GAMEPLAY (opt-in)

Jar	Plugin	Role
yap-vehicles.jar	YaPVehicles	Real vehicle mechanics (not minecarts)
yap-stacker.jar	YaPStacker	PDC mob / item / spawner stacker
yap-gameplay-knobs.jar	YaPGameplayKnobs	Purpur-class mob encyclopedia
yap-skills.jar	YaPSkills	13 RS-style skills + /skills GUI
yap-combat.jar	YaPCombat	Custom PvE combat, gear, food, potions, spells, prayer
yap-crafting.jar	YaPCrafting	Recipes, stations, /sell
yap-mmo-content.jar	YaPMmoContent	Quests v2, bosses, skill areas, hiscores
yap-abilities.jar	YaPAbilities	Config-driven ability engine (230+)
yap-mechanics.jar	YaPMechanics	Stamina, resource nodes, farming

Jar	Plugin	Role
yap-games.jar	YaPGames	Minigames (arenas, queue, duels)
yap-guilds.jar	YaPGuilds	MMO guilds
yap-mmo-bedrock.jar	YaPMmoBedrock	Bedrock MMO forms UI

MMO milestones M0-M7 are shipped — [MMO_PHASES.md](#). Combat skill XP is owned by **YaPCombat** when loaded; YaPSkills provides a vanilla-damage fallback when combat is absent.

6.3 APIs & modules

Nineteen `yap-*-api` jars under `yap-first-party/api/` for soft-depend authors. Fine-tune modules under `modules/` declare `provides` / `requires` and write `FINE_TUNE.txt` pointers at real config knobs — they are **packaging**, not alternate game engines. Games packaging modules (`yap-games-module` , FFA, duels) install with `GAMEPLAY` / `installFineTuneModules` .

6.4 What first-party plugins intentionally replace

Community staple	YaP native
LuckPerms	YaPPerms
EssentialsX (QoL)	YaPEssentials (+ playerdata for data/economy)
CoreProtect	YaPProtect
WorldEdit / FAWE-class ops	YaPWorld (Folia-safe; stock FAWE not used)
WorldGuard	YaPRegions (+ playerdata claims)
TAB / scoreboard	YaPTab
DiscordSRV (MVP)	YaPDiscord
Dynmap-class	YaPMap
PlaceholderAPI	Bundled Clip-compatible engine
Floodgate	YaPFloodgate
Velocity	YaP Link
Via* + Geyser jars	Chassis dual-stack

7. Data plane

7.1 YaPDB

yap-db.jar exposes a shared Hikari pool. Prefer use-shared-yapdb: true in consumers. Setup: ./scripts/db/ensure-db.sh --server-id lobby — [YAPDB.md](#) · [MARIADB.md](#).

7.2 YaPPlayerData

Cross-server inventory / XP / vitals sync, session lock (always on), optional offline /login, economy (/bal /pay + Vault), homes/warps/kits/mail, claims + tax, NPC traders (opt-in).

Default feature flags (September 2026):

Feature	Default
Economy	on
Homes / warps / kits / mail / claims	on
Chest shops (/shop)	on
Auction house (/ah)	on
Jobs	off (keep off when YaPSkills is enabled — avoids double mining payouts)
NPC traders	off (opt-in)

Smoke: . Docs: [PLAYERDATA.md](#).

7.3 Permissions

Native ranks/groups/tracks — [PERMISSIONS.md](#). Soft integration with TAB prefixes and dashboard ranks APIs.

8. YaP Link

First-party **Velocity-class** proxy — **not** a Velocity fork. Phases **0-6 shipped** (0.6.0-phase6): passthrough ping, forced hosts, health failover, chat relay, system chat, Link plugin loader, edge rate limits, metrics hooks, two-backend smoke.

Link plugins (link-plugin.json): chat-bridge, mod-sync, server-selector, tab-bridge, discord. Docs: [YAP_LINK.md](#) · [YAP_LINK_NATIVE.md](#). Stock Velocity remains optional for migration — [VELOCITY.md](#).

9. Networking & crossplay

Path	Notes
Java Edition	Framed Netty; Via-class remapping in chassis (<code>com.yapcore.protocol.via*</code>). Target protocol ~ 776 (26.2). JE floor 1.20.2+ .
Bedrock	UDP path; Geyser-class hub in <code>com.yapcore.crossplay*</code> — no Geyser jar .
Shared listen	Optional same port for JE+BE (<code>shared-listen-port=true</code>).
YaP Link	Optional JE front + optional Bedrock UDP forwarder.
Packs HTTP	Default pack <code>resourcepacks/yapcore-default.zip</code> on <code>:8081</code> .
Publicity	Domain / SRV / nginx + Cloudflare — NETWORKING.md .

Same-machine clients must use `127.0.0.1` (hairpin NAT) — [NGINX_AND_LOCALHOST.md](#).

Phase 4 dual-stack **join DoD is green**; play-depth smoke green; Wave 2 matrix in [CROSSPLAY.md](#).

10. Ops surface

Surface	Detail
Config	<code>config/server.properties</code> + hub under <code>config/</code>
Control GUI	Desktop Swing panel
Web dashboard	Token-auth browser UI <code>:8080</code> — WEB_DASHBOARD.md
Crash dumps	<code>logs/crashes/crash-<ts>-<kind>.log</code>
Release	<code>gradle assembleRelease</code> → <code>build/dist/yapcore-release/{linux,windows}/</code>
Windows	Parity launchers — WINDOWS.md

Dashboard Phase 8 ops tabs are **shipped** (Factions / Guilds / Games interactive; Disasters + MMO reload) — see [WEB_DASHBOARD.md](#).

11. Evaluation methodology

Unit tests (JUnit) cover plugin and API behavior. Operators validate with a local boot: `./scripts/seed-defaults.sh` then `./scripts/start.sh --fg`.

Metrics: region MSPT p99, bridge queue depth, join success by protocol version, HEAVY pool saturation, MariaDB pool health.

12. Threats to validity

- Folia (and YaP-Folia) reject stock single-thread Paper plugins — operator education required.
 - Protocol version sprawl requires continuous registry/packet maintenance.
 - NUMA/ZGC gains are hardware-dependent.
 - Complex JE containers on Bedrock (anvil/smithing/etc.) remain **Out**; Floodgate-only forms are **Limited** — see [CROSSPLAY.md](#).
 - YaP-Folia patch rebase risk when refreshing `UPSTREAM.lock`.
 - Phase 3 spatial edge cases apply only on the **legacy Paper** path.
 - Soft-depend plugin names are **case-sensitive** (`YaPPlayerData`); integrations must match `plugin.yml` `name:` exactly.
-

13. Product status (September 2026)

Area	Status
YaP-Folia product path	Default (<code>folia-jar-source=build</code>)
YapEngine slim chassis	Always on
YaP Link phases 0–6	Shipped (<code>0.6.0-phase6</code>)
Phase 3 Paper spatial	Complete as code — retired as product default
Phase 4 dual-stack join DoD	Green
CORE+NETWORK plugins	Shipped
GAMEPLAY + MMO M0–M7	Shipped (opt-in install)
Ability VFX V1–V4	Shipped — kits + heroes + soak gates (MMO_ABILITY_VFX.md)
PlayerData shops + AH	On by default (jobs remain off)
Fair population MSPT gate	Citeable — fullcite 100 bots; peak –12.4% (<code>shipFc2</code>); re-verify –5.53% (<code>20260904T040935Z</code>); ship knobs disclosed in JSON
Dashboard Phase 8 full tabs	Done (ship) — Factions/Guilds/Games/Disasters/Stacker interactive; Protect restore; Regions flags; MMO reload; YAML-only leftovers documented

Area	Status
Wave 2 Bedrock fidelity	Done (ship) — CROSSPLAY.md matrix; Floodgate forms Limited; anvil/smithing Out
Wave 3 MMO quest objectives	Done (ship) — PLAYTIME/ECONOMY/PLACE/ENCHANT/ANVIL/TALK; silent proxies removed
Wave 4 Discord / Map / PAPI	Done (ship) — event webhooks; map markers; curated local expansions
Wave 5 Access context/temp	Done (ship) — dashboard duration + world/server grants
PAPI eCloud	Intentionally local-only (expansions folder)
Stretch (3D / CFI / full RS / clone mirrors)	Separate plan only — not stubbed (see §13.1)
Stock Paper jars on Folia	Unsupported

Ops notes: [WEB_DASHBOARD.md](#).

13.1 Stretch (deferred — no stub ship)

Item	Why not in Waves 2-5
True BlueMap-style 3D mesh	Flat map + markers is product Map
FAWE CFI / NMS section injection	Folia region-threading conflict
Full RuneScape skill/quest matrix	Wave 3 uses existing YaP systems
DiscordSRV account-link / slash matrix	Event webhooks are product Discord
HelpChat eCloud full mirror	Curated expansions are product PAPI
LuckPerms web pixel-clone	Access context/temp is product ops

Stretch starts only with its own plan and done bars.

14. Conclusion

YaPcore demonstrates a practical decomposition for Minecraft-class servers: **YaP-Folia** owns regionized game tick; **YapEngine** owns a slim edge/I/O chassis with an explicit plugin pool contract; **YaP Link** owns multi-backend routing; and a **first-party plugin + MariaDB data plane** replaces the usual DIY glue stack. Opt-in GAMEPLAY and MMO tiers extend the same Folia-safe patterns for vehicles and RS-style progression.

Future work emphasizes Stretch items (3D map, CFI, full RS) only under a separate plan, hot-region partition soak under load, and continued Folia upstream rebase hygiene.

15. References

1. Minecraft Wiki — *Java Edition protocol*.
2. OpenJDK — *Generational ZGC*.
3. Netty project.
4. PaperMC Folia — regionized threading for Bukkit servers.
5. YapLabs — YaP-Folia patches (`vendor/olia/patches/`), YapEngine chassis notes, YaP Link native suite.
6. YapLabs docs — [YAPCORE_WHITEPAPER.md](#), [YAPCORE_WHITEPAPER.md](#), [MMO_PHASES.md](#).

Appendices

Appendix A — Document map

Audience	Start here
Non-tech readers	Plain English whitepaper , YAPCORE_WHITEPAPER_PLAIN_ENGLISH.md
Operators	QUICK_START.md , QUICK_START.md , PLUGINS.md
Plugin authors	PLUGINS.md , MODULES_AND_API.md , MODULES_AND_API.md
Engine contributors	YAPCORE_WHITEPAPER.md , YAPCORE_WHITEPAPER.md , QUICK_START.md
Network / crossplay	YAP_LINK.md , CROSSPLAY.md , CROSSPLAY.md
Data	YAPDB.md , PLAYERDATA.md , PERMISSIONS.md
MMO	MMO_PHASES.md , MMO_SKILLS.md , MMO_COMBAT.md
Comparison	YAPCORE_WHITEPAPER.md

Appendix B — Quick install

```
# Java 25+
./scripts/build-yap-folia.sh           # → lib/yap-folia-26.2.jar
./scripts/db/ensure-db.sh --server-id lobby
gradle installProductDefaults          # CORE+NETWORK → plugins/
# optional:
gradle installGameplayDefaults         # vehicles + stacker + MMO ...
gradle assembleRelease
./scripts/start.sh --fg
```

```
# multi-backend:
./scripts/start-yap-link.sh
```

Appendix C — Citation

```
@techreport{yapcore2026sixteen,
  title      = {YaPcore: YaP-Folia Game Authority, Slim Edge Chassis, Native Network Stack,
and First-Party Plugin Suite},
  author     = {{YapLabs}},
  institution = {YapLabs},
  year       = {2026},
  month      = sep,
  number     = {YAP-WP-16T-001},
  note       = {Technical whitepaper, YaPcore 0.3}
}
```

Appendix D — Changelog (whitepaper)

Ver	Date	Notes
0.2	Aug 2026	Three-layer architecture; chassis T1-16; Link phase6; Folia product path
0.3	Sep 2026	Comprehensive plugin catalog; MMO M0-M7; data plane (shops/AH defaults); Link/crossplay/ops status; evaluation smokes; honest roadmap gaps

Appendix E — License

YaPcore first-party code is **GNU GPLv3** (or later). See root [LICENSE](#) and [LICENSING.md](#). YaP-Folia derivatives inherit Folia/Paper GPLv3 obligations.